POWER PULSE

Surge Impedance Loading (SIL) Important Formulae

Quick Revision Concepts by BALAJI SIR

1 Surge impedance and SIL

1.
$$Z_c = \sqrt{\frac{L}{C}} = \sqrt{\frac{z}{y}} \ \text{(lossless)}$$

≈ 250–400 Ω for overhead lines

2.
$$SIL = \frac{V_{LL}^2}{Z_c} \ \text{MW} \quad (V_{LL} \text{ in kV},\ Z_c \text{ in } \Omega)$$

3. At SIL: $Q_{gen} = \omega C V^2 = Q_{abs} = \omega L I^2 \ \Rightarrow$ flat voltage profile

4. $P < SIL$: line generates vars, $V_r \uparrow$; $P > SIL$: line absorbs vars, $V_r \downarrow$

2 Wave quantities

5.
$$\beta = \omega\sqrt{LC}, \quad v = \frac{1}{\sqrt{LC}}, \quad \lambda = \frac{v}{f}, \quad \theta = \beta l$$

electrical length

6.
$$P_{max} = \frac{V_s V_r}{Z_c \sin \beta l} = \frac{SIL}{\sin \beta l} \ (V_s = V_r)$$

lossless line

3 Effect of compensation

7.
$$\text{Series } k_{se}: \ Z_c' = Z_c\sqrt{1-k_{se}}, \quad SIL' = \frac{SIL}{\sqrt{1-k_{se}}}$$

SIL increases

8.
$$\text{Shunt } k_{sh}: \ Z_c' = \frac{Z_c}{\sqrt{1-k_{sh}}}, \quad SIL' = SIL\sqrt{1-k_{sh}}$$

SIL decreases

9. Bundling: $L \downarrow,\ C \uparrow \Rightarrow\ Z_c \downarrow,\ SIL \uparrow$

10. SIL $\propto V^2$ (approx. 130 MW at 220 kV, 500 MW at 400 kV, 2300 MW at 765 kV)

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